

Capacitor Charging

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1 Introduction

This document introduces the charging behavior of a resistor-capacitor (RC) network. When a capacitor C is charged through a series R from a constant supply V_0 , the voltage of the capacitor increases according to the condition.

$$V(t) = V_0 \left(1 - e^{-t/RC} \right), \quad (1)$$

where the product RC has units of seconds and is called the time constant $\tau = RC$.